Interactions of Shape-Invariant Waves: Axis-Directional Displacement Transfer, Wave Packet Deformation, and Retroactive Construction of Causality

Author: Noriaki Kihara
Affiliation: WF System Co., Ltd.
ORCID: 0009-0004-6753-4020
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Abstract

Building on the three-mode classification of shape-invariant waves introduced in the companion paper [3] (direction-constrained waves $\kappa = 2$, spherical waves $\kappa = 0$, standing waves $\kappa = 1$), we provide a unified description of interactions between different modes. The fundamental mechanism of interaction is defined as **axis-directional displacement transfer** – the change of a direction-constrained wave's state vector $(\mathbf{k}, \Delta\mathbf{k})$ along the direction of the non-zero axis carried by a spherical wave. Interactions are classified into four fundamental vertex types (absorption, emission, pair annihilation, pair creation), and we prove that under conservation laws, absorption $\Leftrightarrow$ emission and pair annihilation $\Leftrightarrow$ pair creation are reversible. Within this unified framework, we systematically describe all fundamental vertices for the four forces (electromagnetism, strong force, weak force, gravity). Coupling with a standing wave ($\kappa = 1$) modifies the propagation speed ratio $|k_x/k_t|$ of a direction-constrained wave, providing a geometric origin for mass. Under the conserved quantity $\int |\psi|^2 \, dV$, energy transfer deforms the amplitude and width of a wave packet inversely, and this deformation gives a dynamical description of "wave packet collapse." Interactions mediated by spherical waves with $k_t = 0$ carry no intrinsic causal ordering, and the shortest geodesic path and the localization of the wave packet are retroactively constructed from the interaction outcome.

Keywords: shape-invariant wave, fundamental vertex, axis-directional displacement transfer, wave packet deformation, origin of mass, retroactive construction of causality, conservation law

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S1 Introduction and Scope of Claims

This paper provides a unified description of interactions between waves of different modes within the framework of shape-invariant waves constructed in the companion papers [1]–[3], and as consequences derives the origin of mass, the classification of forces, wave packet deformation, and causal structure geometrically.

The claims of this paper are limited to the following six points:

1. All interactions are described uniformly as displacement transfer along the non-zero axes of spherical waves. The difference among types of force reduces solely to the difference in the axis along which the transfer occurs.
2. Interactions are classified into four fundamental vertex types (absorption, emission, pair annihilation, pair creation), and under conservation laws, absorption $\Leftrightarrow$ emission and pair annihilation $\Leftrightarrow$ pair creation are reversible.
3. Coupling with a standing wave ($\kappa = 1$) modifies the speed ratio of a direction-constrained wave, providing a geometric origin for mass.

4. Under the conserved quantity $\int |\psi|^2 \, dV$, energy transfer deterministically deforms the shape of a wave packet, and this deformation corresponds to "wave packet collapse" in observation.
5. Interactions mediated by spherical waves with $k_t = 0$ carry no intrinsic causal ordering, and the path and localization are retroactively constructed from the outcome.
6. The sign of the interaction (attraction / repulsion) between direction-constrained waves is determined geometrically by the relative sign of their t-axis components.

The following elements are **outside the scope of this paper**:

Quantitative derivations:

- Numerical derivation of coupling constants (specific values of α, G_F, α_s, G)
- Quantitative computation of scattering cross sections and decay widths
- Renormalization and scale dependence of running coupling constants
- Rigorous reconstruction of gauge groups in the continuum limit

Details of specific mechanisms:

- Mass acquisition mechanism for $W^{\pm}$ and Z^0 bosons (details of indirect coupling not mediated by spatial axes)
- Mass acquisition mechanism for 1st- and 2nd-generation fermions (details of indirect coupling when the non-zero axis of the standing wave is not directly shared)

Items adopted as assumptions in this paper, with geometric derivation deferred to subsequent papers:

- Geometric derivation of the sign of δk_H (coupling in the cancellation direction; adopted as an assumption in S8.2)
- Geometric derivation of the k_{s_0} sign reversal in W^- coupling (adopted as an additional assumption in S6.5)

Note on physical nomenclature: This paper uses physical names (photon, graviton, electron, flavor, color charge, etc.) assuming correspondence with Table A of [4], but this is for interpretive convenience; the geometric content of each proposition does not depend on the physical identification. All propositions are derived solely from the component structure of wave vectors (κ, k_t, type of axis).

S2 Unified Definition of Interaction

S2.1 Two-Layer Structure

In the tangent bundle $\mathbb{Z}^4 \times \mathbb{Z}^4$ introduced in [2] S8, the state of a direction-constrained wave ($\kappa = 2$) is:

$$(\mathbf{k},\ \Delta\mathbf{k}) \in \mathbb{Z}^4 \times \mathbb{Z}^4$$

- $\mathbf{k}$: **Static label** (type of wave, corresponding to Table A in [4])
- $\Delta\mathbf{k}$: **Dynamic state** (corresponding to energy-momentum)

S2.2 Axis-Directional Displacement Transfer

Definition 2.1 (Axis-directional displacement transfer): When a spherical wave ($\kappa = 0$) interacts with a direction-constrained wave ($\kappa = 2$), the state of the direction-constrained wave changes as follows:

$$(\mathbf{k},\ \Delta\mathbf{k}) \ \longrightarrow\ (\mathbf{k}+\delta\mathbf{k},\ \Delta\mathbf{k}+\delta(\Delta\mathbf{k}))$$

where the displacements $\delta\mathbf{k}$ and $\delta(\Delta\mathbf{k})$ satisfy the following conditions:

(I1) Axiality: The non-zero components of $\delta\mathbf{k}$ and $\delta(\Delta\mathbf{k})$ exist only along the non-zero axis directions in Table A of [4] for the spherical wave. Here, "non-zero axis directions" includes not only continuous displacements along the t and R axes but also discrete label transitions on the Q axis ($Q_i \to Q_j$). That is, for gluons, (I1) requires that the displacement transfer acts only on the Q label change, preserving all other axis components.

(I2) Conservation law: The total sum of tangent bundle states of all waves involved in the interaction is conserved:

$$\sum_i (\mathbf{k}_i + \Delta\mathbf{k}_i) = \text{const}$$

(I3) Wave packet conservation: The conserved quantity $\int |\psi|^2\, dV$ of each wave that exists before and after the interaction is individually conserved. For waves newly created by pair creation (Type D), the initial value of the conserved quantity is set at the time of creation. For waves that are annihilated by pair annihilation (Type C), their conserved quantity is transferred to other waves at the time of annihilation.

S2.3 Static Label Change and Dynamic State Change

Axis-directional displacement transfer encompasses two qualitatively different types of change:

Type of change	Target	Physical meaning	Example
$\delta\mathbf{k} \neq 0$	Static label $\mathbf{k}$	Conversion of wave type	Flavor conversion, color charge exchange
$\delta(\Delta\mathbf{k}) \neq 0$	Dynamic state $\Delta\mathbf{k}$	Change in energy-momentum	Scattering, acceleration

The type of spherical wave determines which (or both) is dominant.

S2.4 Unification of the Four Forces

From the non-zero axis structure of spherical waves in Table A of [4], the four forces are described as applications of the same operation to different axes:

Force	Spherical wave	Non-zero axis	Dominant change
Electromagnetism	$\gamma\ (0,0,0,0,R,0)$	R	$\delta(\Delta\mathbf{k})$ (momentum change)
Strong force	$g\ (0,0,0,0,0,Q_i)$ *	Q	$\delta\mathbf{k}$ (label change)
Weak force	$W^{\pm}\ (0,0,0,\pm t,0,0)$	t	$\delta\mathbf{k}$ (label change)
Gravity	$G\ (0,0,0,\pm t,\pm R,0)$	t, R	$\delta(\Delta\mathbf{k})$ (energy change)

* Whereas γ, $W^{\pm}$, and G are spherical waves with static non-zero components on the t or R axes, g mediates color charge exchange through a spherical wave with discrete label value Q_i on

the Q axis transitioning to a different label value Q_j. That is, the "non-zero axis" of g acts not as a fixed value but as a change in the Q label (detailed in S5).

Proposition 2.1: All four forces are described as axis-directional displacement transfer per Definition 2.1, and the difference among types of force reduces solely to the difference in the axis along which the transfer occurs.

S2.5 R-Axis Discretization Condition (Prerequisite)

This paper treats direction-constrained waves as integer lattice waves on $\mathbb{Z}^4$ (S2.1). For this framework to hold, the curvature radius R must satisfy a number-theoretic constraint.

Theorem 5.2 of prior work [W5] showed that for a 4-dimensional subjective space to be densely regularized with integer values under 5-dimensional central projection, R^2 (in units of the lattice constant) must be a positive integer ($R^2 \in \mathbb{Z}_{>0}$). This is a number-theoretic constraint based on Lagrange's four-square theorem, and if this condition is violated, it becomes impossible to close all phases at integer values.

In [4] S9.8, the discretization $R = \pm n^2 \cdot L_0$ ($n \in \mathbb{N}$, L_0 is the lattice constant) is derived from this condition. That is, the R axis takes not continuous real values but only **discrete values of perfect squares**.

(I4) Consequence of R-axis discreteness: Under the above prerequisite ($R^2 \in \mathbb{Z}_{>0}$), the values that the displacement $\delta(\Delta k)_R$ along the R-axis direction can take are automatically constrained by the requirement that the R values before and after the displacement both satisfy the perfect-square condition. That is, $\delta(\Delta k)_R$ is restricted to the form of the difference of two perfect squares: $n_1^2 - n_2^2 = (n_1 - n_2)(n_1 + n_2)$.

The specific consequences of this discretization condition for individual propositions in this paper — in particular, how it constrains the allowed energy levels in S4 (electromagnetic interaction involving R-axis displacement) and S7 (gravitational interaction involving (t, R)-axis displacement) — remain unresolved and are outside the scope of this paper. This section is placed as a preventive reference point for tracking the consequences of the number-theoretic constraint in subsequent research. For details of the discretization condition, see [W5] S5.5 and [4] S9.8.

S3 Classification and Reversibility of Fundamental Vertices

S3.1 Four Fundamental Vertex Types

The primitive patterns of interaction (fundamental vertices) are classified into the following four types. f denotes a direction-constrained wave ($\kappa = 2$) and B denotes a spherical wave ($\kappa = 0$).

Type A (Absorption): $f + B \longrightarrow f'$

A direction-constrained wave f absorbs a spherical wave B. B ceases to exist as an independent wave, and the $(\mathbf{k}_B + \Delta\mathbf{k}_B)$ carried by B contributes to the state change of f:

$$(\mathbf{k}_f + \Delta\mathbf{k}_f) + (\mathbf{k}_B + \Delta\mathbf{k}_B) = (\mathbf{k}_{f'} + \Delta\mathbf{k}_{f'})$$

Type B (Emission): $f \longrightarrow f' + B$

A direction-constrained wave f emits a spherical wave B. B is newly created as an independent wave, and $(\mathbf{k}_B + \Delta\mathbf{k}_B)$ of B is supplied from the state change of f:

$$(\mathbf{k}_f + \Delta\mathbf{k}_f) = (\mathbf{k}_{f'} + \Delta\mathbf{k}_{f'}) + (\mathbf{k}_B + \Delta\mathbf{k}_B)$$

Type C (Pair Annihilation): $f + \bar{f} \longrightarrow B(s)$

A direction-constrained wave f and its anti-wave $\bar{f}$ annihilate, generating one or more spherical waves B. This is possible because the sum of static labels becomes zero:

$$\mathbf{k}_f + \mathbf{k}_{\bar{f}} = \mathbf{0} \implies \kappa = 2 \to \kappa = 0$$

Type D (Pair Creation): $B(s) \longrightarrow f + \bar{f}$

From one or more spherical waves, a pair $(f, \bar{f})$ of direction-constrained waves is created. A pair of static labels $(\mathbf{k}, -\mathbf{k})$ is created from zero:

$$\kappa = 0 \to \kappa = 2 \impliedby \mathbf{0} \to \mathbf{k}_f + \mathbf{k}_{\bar{f}} = \mathbf{0}$$

S3.2 Reversibility Theorem

Proposition 3.1 (Reversibility): For any process $\mathcal{P}$ satisfying conditions (I1)–(I3), the time-reversed process $\bar{\mathcal{P}}$ also satisfies (I1)–(I3).

Proof: (I1) is a structural condition on axis direction and does not depend on the time direction. (I2) is conservation of a sum, which holds when the initial and final states are exchanged. (I3) is conservation of $\int |\psi|^2 \, dV$ for each wave and does not depend on the direction of the process. □

Corollary 3.1: Type A (absorption) and Type B (emission) are reverse processes of each other, and Type C (pair annihilation) and Type D (pair creation) are reverse processes of each other.

Remark 3.1: When spherical waves with $k_t \neq 0$ ($W^{\pm}$, G) are involved, both the forward and reverse processes are permitted, but due to the temporal asymmetry introduced by $k_t \neq 0$, a causal temporal ordering exists (detailed in S9.5).

S3.3 Composite Processes

More complex interactions are constructed by composing fundamental vertices.

Exchange: Composition of Type B + Type A. f_1 emits B, and f_2 absorbs B:

$$f_1 + f_2 \xrightarrow{B \text{ emission}} f_1' + B + f_2 \xrightarrow{B \text{ absorption}} f_1' + f_2'$$

Scattering: Composition of Type A + Type B. f absorbs B_1 and emits B_2:

$$f + B_1 \xrightarrow{\text{absorption}} f'' \xrightarrow{\text{emission}} f' + B_2$$

Proposition 3.2: All interactions are described as finite compositions of the four fundamental vertex types (Types A–D).

S4 Electromagnetic Interaction

The photon $\gamma = (0, 0, 0, 0, +, 0)$ is a spherical wave with a non-zero component on the R axis. The R axis corresponds to the radial direction of the central projection and acts as a change in the dynamic state $\Delta\mathbf{k}$ on $\mathbb{Z}^4$. By the discretization condition on the R axis (S2.5, (I4)), displacement transfer along the R direction via the photon takes only discrete values. As a candidate consequence of (I4), only discrete allowed values may exist for the energy-momentum carried by the photon.

S4.1 Absorption: $f + \gamma \to f'$

When a direction-constrained wave (electron type) $(k_x, 0, 0, k_t)$ absorbs a photon:

$$(\mathbf{k},\ \Delta\mathbf{k}) + (\mathbf{k}_\gamma,\ \Delta\mathbf{k}_\gamma) \ \longrightarrow\ (\mathbf{k},\ \Delta\mathbf{k} + \delta(\Delta\mathbf{k})_\gamma)$$

The photon ceases to exist as an independent wave, and its energy-momentum is transferred to the dynamic state of the direction-constrained wave. The static label $\mathbf{k}$ does not change (the electron remains an electron).

S4.2 Emission: $f \to f' + \gamma$

By Proposition 3.1 (reversibility), emission is defined as the reverse process of absorption:

$$(\mathbf{k},\ \Delta\mathbf{k}) \ \longrightarrow\ (\mathbf{k},\ \Delta\mathbf{k} - \delta(\Delta\mathbf{k})_\gamma) + (\mathbf{k}_\gamma,\ \delta(\Delta\mathbf{k})_\gamma)$$

A portion of the dynamic state of the direction-constrained wave separates, and a new photon is created as an independent wave. The static label $\mathbf{k}$ is conserved.

Proposition 4.1: Interactions with the photon (both absorption and emission) preserve the static label of the direction-constrained wave and change only the dynamic state $\Delta\mathbf{k}$.

S4.3 Exchange and Attraction/Repulsion

The electromagnetic interaction between two direction-constrained waves ψ_1, ψ_2 is described as photon exchange (S3.3):

$$\psi_1 \xrightarrow{\gamma \text{ emission}} \psi_1' + \gamma \xrightarrow{\gamma \text{ absorption}} \psi_1' + \psi_2'$$

By conservation law (I2):

$$\Delta\mathbf{k}_1 + \Delta\mathbf{k}_2 = \Delta\mathbf{k}_1' + \Delta\mathbf{k}_2'$$

Consider the action of $\delta(\Delta\mathbf{k})_\gamma$ on the t-axis components in photon exchange. The photon has a non-zero component on the R axis ((I1)), but the signs of $k_{t,1}$ and $k_{t,2}$ of the two direction-constrained waves involved in the interaction constrain the direction of momentum change under conservation law (I2). In the process where ψ_1 emits a photon and ψ_2 absorbs it, $\delta(\Delta\mathbf{k})_\gamma$ is transferred from ψ_1 to ψ_2. When $k_{t,1}$ and $k_{t,2}$ have the same sign, the relative momentum change under (I2) conservation is directed to separate the two waves (repulsion); when the signs are opposite, it is directed to bring them together (attraction).

Concrete example: When $k_{t,1}, k_{t,2}$ are both positive, ψ_1 emits a photon with R-axis displacement $\delta(\Delta k)_R$ and ψ_2 absorbs it; by (I2), $\Delta k_{R,1}$ decreases by $\delta(\Delta k)_R$ while $\Delta k_{R,2}$ increases by

the same amount. Two waves with the same sign of k_t have co-directional temporal evolution in the radial structure of the central projection ([5] S5), and this momentum redistribution acts spatially to separate the two waves (repulsion). For opposite signs, the contra-directional temporal evolution causes the redistribution to act in the direction of attraction.

The t-axis component k_t corresponds to the charge structure in Table A of [4]:

k_t of ψ_1	k_t of ψ_2	Sign product	Interaction
$+$	$+$	$+1$	Repulsion
$+$	$-$	-1	Attraction
$-$	$-$	$+1$	Repulsion

Proposition 4.2: The sign of the electromagnetic interaction between two direction-constrained waves is determined geometrically by the relative sign $\mathrm{sign}(k_{t,1} \cdot k_{t,2})$ of their t-axis components. The specific correspondence "same sign = repulsion, opposite sign = attraction" depends on the interpretive layer of the correspondence between k_t and charge sign in Table A of [4]; the geometric content of the proposition is that "the relative sign uniquely determines the directionality of the interaction."

S4.4 Pair Annihilation: $f + \bar{f} \to \gamma(s)$

A direction-constrained wave and its antiparticle are in a relationship where all components have opposite signs (C transformation in [3] S6.5):

$$\psi : (k_x, 0, 0, k_t) \quad \longleftrightarrow \quad \bar{\psi} : (-k_x, 0, 0, -k_t)$$

When both interact in the same lattice region, the sum of static labels becomes zero:

$$(k_x, 0, 0, k_t) + (-k_x, 0, 0, -k_t) = (0, 0, 0, 0)$$

A pair of $\kappa = 2$ waves transitions to a $\kappa = 0$ state. By conservation law (I2), the energy of the vanished static labels is emitted as spherical waves (photons).

Proposition 4.3: Pair annihilation of a direction-constrained wave and an anti-direction-constrained wave is a process in which a pair of $\kappa = 2$ waves is converted to $\kappa = 0$ spherical waves, made possible by the sum of static labels becoming zero.

S4.5 Pair Creation: $\gamma(s) \to f + \bar{f}$

By Proposition 3.1 (reversibility), pair creation is defined as the reverse process of pair annihilation. From the energy of a spherical wave, a pair of direction-constrained waves with static labels $(\mathbf{k}, -\mathbf{k})$ is created:

$$\kappa = 0 \;\longrightarrow\; \kappa = 2 : \quad (0, 0, 0, 0) \;\longrightarrow\; (k_x, 0, 0, k_t) + (-k_x, 0, 0, -k_t)$$

By conservation law (I2), the dynamic state $\Delta\mathbf{k}_\gamma$ of the spherical wave is distributed to the created pair of direction-constrained waves.

Proposition 4.4: Pair creation is the reverse process of pair annihilation; a pair of $\kappa = 2$ waves is created from the energy of a spherical wave. The static labels of the created direction-constrained waves are restricted to antisymmetric $(\mathbf{k}, -\mathbf{k})$.

S5 Strong Interaction and Confinement

S5.1 Q-Axis Transition and the 2-Bit Structure of Color Charge

The gluon $g = (0, 0, 0, 0, 0, Q_i \to Q_j)$ is defined as a transition on the Q axis. By condition (I1), the displacement transfer acts only in the Q-axis direction:

$$\delta \mathbf{k}_g : \quad Q \longrightarrow Q'$$

The spatial, temporal, and R components of the direction-constrained wave do not change. Only the discrete label Q changes.

In Table A of [4], Q is a 4-valued discrete label encoded as 2 bits (c_2, c_3):

$$Q = 2c_2 + c_3 \in \{0, 1, 2, 3\} \quad (c_i \in \{0, 1\})$$

- $Q = 0$ ($c_2c_3 = 00$): Zero Q-axis component, i.e., orthogonal to the Q axis (no color charge, lepton type)
- $Q \in \{1, 2, 3\}$ ($c_2c_3 \in \{01, 10, 11\}$): Three colors of color charge (quark type)

Weak isospin is not encoded in the Q axis but is independently defined as the sign of the t axis (sign(k_t); see S6). The Q axis and the t axis are mutually orthogonal, independent degrees of freedom.

The Q transition of gluons is a change among non-zero Q values ($\{1, 2, 3\}$), corresponding to color charge exchange. This yields $3 \times 3 - 1 = 8$ independent gluons (the -1 corresponds to the exclusion of the $Q = 0$ color-neutral mode).

S5.2 Absorption: $q + g \to q'$

When a quark-type direction-constrained wave with color charge $(c_2c_3) = (01)$ (red) absorbs gluon $g_{01\to10}$:

$$(k_x, 0, 0, k_t, 0, Q_{01}) + g_{01\to10} \longrightarrow (k_x, 0, 0, k_t, 0, Q_{10})$$

The gluon ceases to exist as an independent wave, and the color charge of the quark changes from red (01) to green (10). The spatial, temporal, and R components are conserved.

S5.3 Emission: $q \to q' + g$

By Proposition 3.1 (reversibility), emission is defined as the reverse process of absorption:

$$(k_x, 0, 0, k_t, 0, Q_{10}) \longrightarrow (k_x, 0, 0, k_t, 0, Q_{01}) + g_{10\to01}$$

The color charge of the quark changes, and a gluon carrying the difference is newly created.

Proposition 5.1: Interactions with gluons (both absorption and emission) change only the Q label (color charge) of the direction-constrained wave, preserving the spatial, temporal, R components, and dynamic state.

S5.4 Pair Annihilation and Pair Creation: $q + \bar{q} \leftrightarrow g(s)$

In a quark-antiquark pair, the spatial axes (x, y, z) have inverted signs, while the Q value (color charge) and the sign of the t axis remain unchanged ([4] S9.9). Therefore, the antiquark carries the same color charge (c_2c_3) as the corresponding quark, and the bitwise composition (XOR) of color charges reduces to zero:

$$(c_2c_3) \oplus (c_2c_3) = (00)$$

Pair annihilation occurs, and the $\kappa = 2$ wave pair is converted to $\kappa = 0$ gluons. As the reverse process (pair creation), a quark-antiquark pair with the same color (c_2c_3) is generated from gluon energy:

$$q(c_2c_3) + \bar{q}(c_2c_3) \;\longleftrightarrow\; g(s)$$

Proposition 5.2: Quark-antiquark pairs carrying the same color charge can undergo pair annihilation and pair creation via gluons, as the XOR composition of their color charges reduces to zero.

S5.5 Confinement Condition

Within the framework of subspace central projection introduced in [3] S8, to form an independent subjective space, a collection of waves must be **color-neutral**.

Definition 5.1 (Color neutrality condition): A set $\{\psi_i\}$ of direction-constrained waves is color-neutral if the composition of c_2c_3 bits reduces to $(0, 0)$. Here, bit composition is defined as bitwise exclusive OR (XOR, addition over $\mathbb{F}_2$):

$$(a_1, a_2) \oplus (b_1, b_2) = (a_1 \oplus b_1,\ a_2 \oplus b_2) \quad (\oplus : \text{mod 2 addition})$$

Realizations of color neutrality:

Configuration	Color combination	Name
3 waves	$(01) \oplus (10) \oplus (11) = (00)$	Baryon type
2 waves	$(c_2c_3) \oplus (c_2c_3) = (00)$	Meson type
1 wave	$(c_2c_3) = (00)$	Lepton type

Remark 5.1: In the meson type, the quark and antiquark carry the same color (c_2c_3) (S5.4), so color neutrality is achieved by the XOR composition of two identical colors. In standard QCD, color neutrality is described as the annihilation of color and anti-color (SU(3): $\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$), whereas in this framework, XOR algebra over $\mathbb{F}_2^2$ provides the corresponding operation. The algebraic structures differ, but the result – color neutrality – is identical.

Proposition 5.3: A set of direction-constrained waves that does not satisfy the color neutrality condition cannot form an independent subjective space and is confined within a color-neutral set.

S5.6 Asymptotic Freedom

Inside a color-neutral set (inside the shared subjective space), the curvature of the subspace central projection is small, and direction-constrained waves move almost freely. As one approaches the boundary of the set, the projection curvature increases, making it impossible to cross the boundary.

Proposition 5.4: The interior of a color-neutral set has small projection curvature (asymptotic freedom), while the curvature diverges at the boundary (confinement).

S6 Weak Interaction and Flavor Conversion

S6.1 t-Axis Displacement

$W^{\pm} = (0, 0, 0, \pm t, 0, 0)$ is a spherical wave with a non-zero component on the t axis. By condition (I1):

$$\delta \mathbf{k}_W : \quad k_t \longrightarrow k'_t = k_t + \delta k_t$$

The change in the t-axis component corresponds to the conversion between up-type ($k_t > 0$) and down-type ($k_t < 0$) in Table A of [4]. Weak isospin is encoded as the sign of the t axis, and $W^{\pm}$ inverts this sign. The Q axis (color charge) is an independent degree of freedom orthogonal to the t axis and does not change under interaction with $W^{\pm}$.

The t-axis displacement by $W^{\pm}$ acts as a sign inversion when k_t is non-zero in both the initial and final states (quark system, S6.2–6.3). When the transition involves $k_t = 0$ (lepton system), axis reconfiguration accompanying $\kappa = 2$ preservation occurs (S6.7).

S6.2 Absorption: $f + W \to f'$

Transformation of a direction-constrained wave by W^+ absorption:

$$\underbrace{(k_x, 0, 0, -k_t, 0, Q)}_{\text{down-type}} + \underbrace{(0, 0, 0, +, 0, 0)}_{W^+} \longrightarrow \underbrace{(k_x, 0, 0, +k_t, 0, Q)}_{\text{up-type}}$$

W^+ ceases to exist as an independent wave, and the sign of the t axis of the direction-constrained wave is inverted ($-k_t \to +k_t$). Q (color charge) is conserved.

S6.3 Emission: $f \to f' + W$

By Proposition 3.1 (reversibility), emission is defined as the reverse process of absorption:

$$\underbrace{(k_x, 0, 0, +k_t, 0, Q)}_{\text{up-type}} \longrightarrow \underbrace{(k_x, 0, 0, -k_t, 0, Q)}_{\text{down-type}} + \underbrace{(0, 0, 0, +, 0, 0)}_{W^+}$$

The flavor of the direction-constrained wave changes from up-type to down-type, and W^+ is newly created as an independent wave. By conservation law (I2), the change in the t-axis component is emitted as W^+.

Orthogonality of the Q axis and the t axis: In the revised version of [4], weak isospin is not encoded in the Q axis but is independently defined as the sign $\text{sign}(k_t)$ of the t axis. The Q axis encodes only color charge as a 2-bit structure (S5.1) and is orthogonal to the t axis. Therefore,

interactions with $W^{\pm}$ change only the t-axis component, and Q (color charge) is conserved. (I1) requires displacement transfer only along the non-zero axis of the spherical wave (the t axis), and the invariance of the Q axis is a direct consequence of this condition.

Proposition 6.1: Interactions with $W^{\pm}$ (both absorption and emission) change only the t-axis component (flavor) of the direction-constrained wave. The Q axis (color charge) is an independent degree of freedom orthogonal to the t axis and is conserved under interactions with $W^{\pm}$.

S6.4 W Decay and Inverse Decay

The decay of $W^{\pm}$ itself is a special case of pair creation (Type D):

$$W^{+} \longrightarrow f_u + \bar{f}_d$$

W^{+} is annihilated, and an up-type direction-constrained wave f_u and a down-type anti-direction-constrained wave $\bar{f}_d$ are created. By conservation law (I2), the t-axis component and dynamic state of W^{+} are distributed to the created wave pair.

Reverse process (inverse decay):

$$f_u + \bar{f}_d \longrightarrow W^{+}$$

A direction-constrained wave and an anti-direction-constrained wave are annihilated, and W^{+} is created. This corresponds to the t-axis version of pair annihilation (Type C).

Proposition 6.2: W decay is the t-axis version of Type D, and inverse decay is the t-axis version of Type C. Both are causal processes with $k_t \neq 0$, and decay and inverse decay are temporally distinguishable.

S6.5 Chirality Selection Rule

For the chirality $\chi = \mathrm{sign}(k_{s_0} \cdot k_t)$ (k_{s_0}: non-zero spatial component) defined in Definition 6.1 of [3], the t-axis structure of $W^{\pm}$ imposes a coupling condition with specific chirality.

$W^{\pm} = (0, 0, 0, \pm t, 0, 0)$ has phase structure only on the t axis. We derive the coupling condition by directly computing the chirality of the initial state in each case.

Case of W^{+} absorption: $W^{+} + \psi_d \to \psi_u$. The initial-state down-type fermion is $\psi_d = (k_{s_0}, 0, 0, -|k_t|)$ ($k_t < 0$). For coupling to occur, the t component of W^{+} ($+$) and the t component of the direction-constrained wave ($-|k_t|$) must have opposite signs. Therefore, the initial state is restricted to $k_t < 0$. The chirality of the initial state is:

$$\chi(\psi_d) = \mathrm{sign}(k_{s_0} \cdot k_t) = \mathrm{sign}(k_{s_0} \cdot (-|k_t|)) = -\mathrm{sign}(k_{s_0})$$

Since $k_{s_0} > 0$ (convention of Table A in [4]), $\chi(\psi_d) = -1$.

Case of W^{-} absorption: $W^{-} + \psi_u \to \psi_d$. The initial-state up-type fermion is $\psi_u = (k_{s_0}, 0, 0, +|k_t|)$ ($k_t > 0$). The condition that the t component of W^{-} ($-$) and the t component of the direction-constrained wave ($+|k_t|$) have opposite signs is satisfied. The chirality of the initial state is:

$$\chi(\psi_u) = \mathrm{sign}(k_{s_0} \cdot k_t) = \mathrm{sign}(k_{s_0} \cdot (+|k_t|)) = \mathrm{sign}(k_{s_0})$$

If $k_{s_0} > 0$ (convention of Table A in [4]), then $\chi(\psi_u) = +1$ (right-handed), and $\chi = -1$ is not directly derived as in the W^+ case.

Additional assumption (k_{s_0} sign reversal): W^+ and W^- form an antisymmetric pair on the t axis ($+t$, $-t$). We assume that, corresponding to this t-axis antisymmetry, the wave vectors of the coupling partners are also inverted in the (k_{s_0}, k_t) plane: W^+ couples to the region $(k_{s_0} > 0,\ k_t < 0)$, while W^- couples to the region $(k_{s_0} < 0,\ k_t > 0)$. Under this assumption, the coupling partner of W^- is restricted to $k_{s_0} < 0$, yielding:

$$\chi = \mathrm{sign}((-) \cdot (+)) = -1$$

The geometric derivation of this k_{s_0} sign reversal is outside the scope of this paper (see S1).

Proposition 6.3: For W^+, the opposite-sign condition on t-axis phase directly derives that the chirality of the initial state is $\chi = -1$ (left-handed). For W^-, $\chi = -1$ likewise holds under the additional assumption of k_{s_0} sign reversal based on t-axis antisymmetry. Right-handed ($\chi = +1$) direction-constrained waves do not couple to $W^\pm$.

S6.6 CP Violation

As shown in Proposition 6.4 of [3], CP symmetry can only be violated in interactions with $k_t \neq 0$. $W^\pm$ has $k_t = \pm 1 \neq 0$ and is the only one among the four types of spherical waves with $k_t \neq 0$.

Proposition 6.4: Only $W^\pm$, which has a non-zero component on the t axis, mediates CP symmetry violation. This is a direct consequence of $k_t \neq 0$ introducing temporal asymmetry.

S6.7 Axis Reconfiguration in the Lepton System

The $W^\pm$ absorption and emission described in S6.2–6.3 treated the sign inversion of k_t (where $k_t \neq 0$ in both initial and final states). This is the typical process for the quark system ($Q \neq 0$). In the lepton system ($Q = 0$), the axis configurations of charged leptons and neutrinos differ qualitatively, so the action of $W^\pm$ involves axis reconfiguration.

In Table A of [4], fermions ($\kappa = 2$) with $Q = 0$ (orthogonal to the Q axis) have the following two types of axis configurations:

Type	Non-zero axes	k_t	Example
Charged lepton	1 spatial axis + t	$\neq 0$	$e^- : (k_x, 0, 0, k_t, 0, 0)$
Neutrino	2 spatial axes	$= 0$	$\nu : (k_{s_1}, 0, k_{s_2}, 0, 0, 0)$

Charged lepton → neutrino conversion by W^+ absorption:

$$\underbrace{(k_x, 0, 0, k_t, 0, 0)}_{\substack{\text{charged lepton} \\ \kappa=2:\ (s,t)}} + \underbrace{(0, 0, 0, +, 0, 0)}_{W^+} \longrightarrow \underbrace{(k_x, 0, k_{s_2}, 0, 0, 0)}_{\substack{\text{neutrino} \\ \kappa=2:\ (s,s)}}$$

This process cannot be fully described by (I1) alone. (I1) requires displacement along the non-zero axis of W^+ (the t axis) and permits the transition $k_t \to 0$. However, the transition to $k_t = 0$ reduces the number of non-zero axes from $\kappa = 2$ to $\kappa = 1$, so the stability condition for $\kappa = 2$ (Theorem 5.1 of [3]) demands compensatory activation of a spatial axis.

That is, the action of $W^{\pm}$ on the lepton system is determined as the simultaneous satisfaction of two conditions:

1. **(I1)**: Displacement along the t axis ($k_t \to 0$ or $0 \to k_t$)
2. $\kappa = 2$ **preservation**: The fermion condition maintains the number of non-zero axes at $\kappa = 2$

Reverse process (neutrino → charged lepton conversion):

$$\underbrace{(k_x, 0, k_{s_2}, 0, 0, 0)}_{\substack{\text{neutrino} \\ \kappa=2:\ (s,s)}} + \underbrace{(0, 0, 0, -, 0, 0)}_{W^-} \longrightarrow \underbrace{(k_x, 0, 0, k_t, 0, 0)}_{\substack{\text{charged lepton} \\ \kappa=2:\ (s,t)}}$$

W^- supplies a t-axis component ($0 \to k_t$), and $\kappa = 2$ preservation requires that one of the spatial axes (k_{s_2}) becomes deactivated.

Proposition 6.5: The action of $W^{\pm}$ on the lepton system is described as the simultaneous satisfaction of t-axis displacement (I1) and $\kappa = 2$ preservation. The conversion between charged leptons (1 spatial axis + t) and neutrinos (2 spatial axes) preserves $\kappa = 2$ through the interchange of t-axis and spatial-axis components.

Remark 6.1: The choice of which spatial axis is activated (or deactivated) in the axis reconfiguration is not uniquely determined by $\kappa = 2$ preservation alone. For example, from $e^- : (k_x, 0, 0, k_t)$, a neutrino generated by W^+ absorption could be either $(k_x, k_y, 0, 0)$ or $(k_x, 0, k_z, 0)$, both satisfying $\kappa = 2$. This degree of freedom may constitute the geometric origin of neutrino mixing (a structure corresponding to the PMNS matrix), but its elucidation is outside the scope of this paper.

S7 Gravitational Interaction

S7.1 Displacement Transfer by the Graviton

The graviton $G = (0, 0, 0, \pm t, \pm R, 0)$ has non-zero components on both the t and R axes. By condition (I1), the displacement transfer acts in both the t and R directions, with the R direction subject to the discretization condition (I4) (S2.5):

$$\delta(\Delta\mathbf{k})_G = (\delta(\Delta k)_t,\ \delta(\Delta k)_R)$$

This means simultaneous changes in energy (t component) and spatial configuration (R component). As a candidate consequence of (I4), only discrete allowed values may exist for the energy-momentum carried by the graviton along the R direction.

S7.2 Absorption: $f + G \to f'$

When a direction-constrained wave absorbs a graviton:

$$(\mathbf{k},\ \Delta\mathbf{k}) + (\mathbf{k}_G,\ \Delta\mathbf{k}_G) \ \longrightarrow\ (\mathbf{k},\ \Delta\mathbf{k} + \delta(\Delta\mathbf{k})_G)$$

The graviton ceases to exist as an independent wave, and its energy-momentum is transferred to the dynamic state of the direction-constrained wave. The simultaneous change in both the t and R directions distinguishes this from the photon (R only).

S7.3 Emission: $f \to f' + G$

By Proposition 3.1 (reversibility), emission is defined as the reverse process of absorption:

$$(\mathbf{k},\ \Delta\mathbf{k}) \ \longrightarrow\ (\mathbf{k},\ \Delta\mathbf{k} - \delta(\Delta\mathbf{k})_G) + (\mathbf{k}_G,\ \delta(\Delta\mathbf{k})_G)$$

A portion of the dynamic state of the direction-constrained wave separates, and a new graviton is created as an independent wave.

Proposition 7.1: Interactions with the graviton (both absorption and emission) change only the dynamic state $\Delta\mathbf{k}$ and do not act on the static label $\mathbf{k}$.

S7.4 Universal Coupling

We restate the description of acceleration introduced in [3] S8. The second-order difference of the subspace central projection determines the acceleration:

$$\Delta^2\mathbf{k}(\mathbf{k}_t) = D^2\pi(\mathbf{k}_t) \cdot \Delta\mathbf{k}_t$$

Graviton exchange is the discrete realization of this $\Delta^2\mathbf{k}$. Since the graviton acts on the dynamic state $\Delta\mathbf{k}$:

- It is independent of the static label $\mathbf{k}$ (regardless of wave type)
- It acts on all waves with $\Delta\mathbf{k} \neq 0$ (all waves carrying energy)

Proposition 7.2: All shape-invariant waves with $\Delta\mathbf{k} \neq 0$ couple to the graviton (universal coupling).

Remark 7.1: Pair annihilation ($f + \bar{f} \to G$) and pair creation ($G \to f + \bar{f}$) via gravity are permitted in principle under conservation law (I2). However, since the graviton has $k_t \neq 0$, these are causal processes, and since coupling is mediated only through the dynamic state, the coupling strength is remarkably weak compared to other forces. Quantitative evaluation of these processes is outside the scope of this paper.

S8 Origin of Mass: Coupling with Standing Waves

S8.1 Geometric Structure of Coupling

The standing wave mode ($\kappa = 1$, [3] S5) fills the entire subjective space S^3 as a fundamental mode along one spatial axis. Direction-constrained waves ($\kappa = 2$) propagate through this standing wave background.

Remark 8.1: Higgs coupling is outside the scope of the fundamental vertex classification of S3 (Types A–D, which involve spherical waves with $\kappa = 0$). The standing wave ($\kappa = 1$) is not a spherical wave ($\kappa = 0$), and its coupling with direction-constrained waves is described independently in this section as a modification of the speed ratio, not through fundamental vertices.

In Table A of [4], the non-zero axis of the Higgs boson is assigned to the z axis ($(0, 0, +, 0, 0, 0)$). The z axis corresponds to the 3rd generation in the generation labels (generation 1: x, generation 2: y, generation 3: z, [4] S9.9). The conclusions of S8 of this paper (direction of the generation mass hierarchy) depend on this correspondence between generation numbers and spatial axes, and presuppose that the axis assignment of H^0 in Table A of [4] and the generation label

assignment are mutually consistent. With this assignment, the standing wave is expressed as $\mathbf{k}_H = (0, 0, k_H, 0)$.

A 3rd-generation direction-constrained wave $\mathbf{k} = (0, 0, k_z, k_t)$ and the standing wave $\mathbf{k}_H = (0, 0, k_H, 0)$ directly share the spatial z axis. On this shared axis, the phase structures of the waves overlap, producing the strongest coupling. 1st-generation ($\mathbf{k} = (k_x, 0, 0, k_t)$) and 2nd-generation ($\mathbf{k} = (0, k_y, 0, k_t)$) direction-constrained waves do not share the z axis with the standing wave, so they lack direct coupling and only indirect coupling through the global topological structure of S^3 is possible.

S8.2 Modification of the Speed Ratio

The propagation speed of a direction-constrained wave without coupling is $v = k_{s_0}/k_t$ (k_{s_0}: non-zero spatial component). Coupling with the standing wave affects the phase structure along the shared spatial axis direction, modifying the effective spatial component:

$$k_{s_0} \longrightarrow k_{s_0}^{\text{eff}} = k_{s_0} - \delta k_H$$

where $\delta k_H > 0$ is proportional to the coupling strength with the standing wave. **Sign assumption**: The positivity of δk_H (the direction of reducing the effective spatial component) is adopted as an assumption in this paper. The physical motivation is a picture in which the background phase structure of the standing wave overlaps in antiphase with the spatial phase of the direction-constrained wave and couples in the cancellation direction, but the geometric derivation of this sign choice is deferred to a subsequent paper (see S1). The propagation speed after coupling is:

$$v^{\text{eff}} = \frac{k_{s_0}^{\text{eff}}}{k_t} = \frac{k_{s_0} - \delta k_H}{k_t} < \frac{k_{s_0}}{k_t} = v$$

The speed decreases – this is the **geometric origin of mass**.

Proposition 8.1: Coupling with a standing wave ($\kappa = 1$) reduces the effective spatial component of a direction-constrained wave, limiting the propagation speed to $v < c$. This speed limitation is observed as mass.

S8.3 Waves That Do Not Couple

Spherical waves ($\kappa_s = 0$) have no spatial components. The standing wave has phase structure along one spatial axis, but spherical waves have no spatial axis to share.

Proposition 8.2: Spherical waves with $\kappa_s = 0$ have no coupling axis with the standing wave, yielding $\delta k_H = 0$. The speed remains $v = c$ without modification.

This is consistent with the gauge bosons with $\kappa_s = 0$ (γ, g) having zero mass in Table A of [4]. $W^{\pm}$ and Z^0 have $\kappa_s = 0$ and therefore lack direct coupling through spatial axes as described by Proposition 8.2, but they may acquire mass through an indirect standing wave coupling mechanism via the t or R axis. Proposition 8.2 addresses only coupling through direct spatial axis sharing; the details of indirect coupling mechanisms via the t or R axis are outside the scope of this paper.

S8.4 Mass Hierarchy

The coupling strength δk_H depends on the overlap integral along the spatial axis between the direction-constrained wave and the standing wave. With the standing wave's non-zero

axis assigned to z (the 3rd-generation axis) in S8.1, the ordering of coupling strengths among generations is structurally determined:

Generation	Direction-constrained wave	Axis sharing with standing wave	Coupling strength δk_H
3rd	$(0, 0, k_z, k_t)$	Direct sharing of z axis	Maximum
1st / 2nd	$(k_x, 0, 0, k_t)$ / $(0, k_y, 0, k_t)$	No shared axis (indirect coupling)	Small (mechanism unresolved)

The 3rd generation directly shares the spatial axis with the standing wave, yielding the maximum δk_H and acquiring the largest mass. Empirically, the 3rd generation is the heaviest ($m_t \gg m_c \gg m_u$), which is consistent with this structural prediction.

The 1st and 2nd generations do not share the z axis and therefore have only indirect coupling through the global topological structure of S^3. However, since the x and y axes are locally equivalent under SO(3) symmetry, the difference $m_2 \neq m_1$ cannot be derived within the framework of this paper. The origin of the mass difference between the 1st and 2nd generations depends on an additional mechanism that breaks this SO(3) symmetry (the detailed structure of indirect coupling), but its elucidation is outside the scope of this paper (see S1).

What this paper structurally claims is only that "$m_3 \gg m_{1,2}$" (the 3rd generation is overwhelmingly heavier than the 1st and 2nd generations).

S9 Wave Packet Deformation and Retroactive Construction of Causality

S9.1 Conservation-Law Description of Wave Packet Deformation

In the interactions of S4–S7, when a direction-constrained wave receives energy, under the conserved quantity $\int |\psi|^2 \, dV = E_0$ ((C1) from [1] S5, [2] S5), the amplitude A and characteristic width w of the wave packet satisfy:

$$A^2 \cdot w^d = E_0 = \text{const} \quad (d = 3 : \text{spatial dimension of } S^3)$$

A wave packet that receives energy:

$$\Delta E > 0 \;\Rightarrow\; A \uparrow,\; w \downarrow \quad \text{(amplitude increase, width decrease)}$$

$$\Delta E < 0 \;\Rightarrow\; A \downarrow,\; w \uparrow \quad \text{(amplitude decrease, width increase)}$$

Proposition 9.1: Under the conserved quantity $\int |\psi|^2 \, dV$, energy transfer deforms the amplitude and width of a wave packet inversely. Energy increase is accompanied by localization (width decrease) of the wave packet.

S9.2 Wave Packet Deformation of Spherical Waves

Spherical waves ($\kappa = 0$) also couple to the graviton (Proposition 7.2: coupling occurs if $\Delta \mathbf{k} \neq 0$). When a spherical wave receives energy:

- The amplitude of the spherical wave increases and its spatial spread decreases

- In the limit, the spherical wave becomes point-like localized

This corresponds to the geometric description of gravitational lensing. When a spherical wave (photon) passes through the gravitational field of a massive body, the wave packet is deformed through interaction with gravitons, bending the propagation path.

S9.3 Dynamical Description of "Observation"

In conventional quantum mechanics, "wave packet collapse through observation" is introduced axiomatically. In this framework:

1. A direction-constrained wave ($\kappa = 2$) interacts with a spherical wave ($\kappa = 0$)
2. Energy transfer increases the amplitude of the wave packet
3. By conservation law (I3) and Proposition 9.1, the width decreases (localization)
4. A sufficiently localized wave packet is described as "detected as a particle"

"Observation" is not a special operation but merely one example of wave packet deformation through interaction. No external "observer" is required, and wave packet deformation is a deterministic process following conservation laws.

Proposition 9.2: Wave packet collapse is a conservation-law-governed wave packet deformation through interaction, and there is no need to postulate a special process called "observation."

S9.4 Retroactive Construction of Causality

As shown in Proposition 4.2 of [3], spherical waves with $k_t = 0$ are time-reversal symmetric and carry no intrinsic causal ordering. The "deterministic" of Proposition 9.2 and the "retroactive" below are not contradictory. They belong to different layers:

- **Deterministic**: Conservation laws (I2)(I3) hold strictly, and the result of wave packet deformation is uniquely determined
- **Retroactive**: Due to $k_t = 0$, causal ordering (the temporal order of "cause $\rightarrow$ effect") is not intrinsic, so which event is the "cause" and which is the "effect" is indeterminate

That is, the structure "conservation laws are strictly followed, but causal ordering is absent" arises naturally from the time-reversal symmetry of $k_t = 0$. Combining this property with wave packet deformation:

(R1) Retroactive determination of path: A spherical wave ($k_t = 0$) propagates isotropically in all directions (Proposition 4.1 of [3]). Viewed from the result of the interaction, the spherical wave **appears to have** propagated along the shortest geodesic. Since causal ordering does not exist, the result retroactively determines the path.

(R2) Retroactive determination of localization: Wave packet deformation is a deterministic process following conservation laws (Proposition 9.2). However, since $k_t = 0$ eliminates causal ordering, "the interaction occurred and therefore localization happened" and "localization happened and therefore the interaction appears to have occurred" are indistinguishable.

(R3) Principle of least action: As a synthesis of (R1) and (R2), the variational principle of natural law (the principle of least action) is not an axiom but is derived as a consequence of the causal symmetry of interactions mediated by spherical waves with $k_t = 0$.

Proposition 9.3: In interactions mediated by spherical waves with $k_t = 0$, (a) the shortest propagation path, (b) wave packet localization, and (c) the variational principle are all retroactively constructed from the absence of causal ordering.

S9.5 The Case of $k_t \neq 0$

Interactions mediated by $W^{\pm}$ ($k_t \neq 0$) and G ($k_t \neq 0$) are not causally symmetric. (R1)–(R3) do not apply:

- The path is causally determined (emission first, absorption later)
- Time-reversal symmetry is broken
- CP non-conservation becomes possible (Proposition 6.4, $W^{\pm}$ only)

k_t of spherical wave	Causal ordering	Path selection	CP	Corresponding force
$k_t = 0$	None	Retroactive (geodesic)	Conserved	Electromagnetism, strong force
$k_t \neq 0$	Present	Causal	May be violated	Weak force, gravity

Combined with Remark 3.1 of S3.2: when $k_t = 0$, absorption and emission are temporally indistinguishable (the reversibility of S3 is completely symmetric), whereas when $k_t \neq 0$, even though they are reversible, they are temporally distinguishable.

S10 Summary

The main results of this paper are summarized:

1. **Unified interaction mechanism**: All forces are described as displacement transfer (I1)–(I3) along non-zero axes of spherical waves, and the type of force reduces solely to the difference in axes (S2).

2. **Fundamental vertices and reversibility**: Interactions are classified into four fundamental vertex types (absorption, emission, pair annihilation, pair creation), and under conservation laws, absorption $\Leftrightarrow$ emission and pair annihilation $\Leftrightarrow$ pair creation are reversible (S3).

3. **Electromagnetic interaction**: The photon acts on the R axis and changes only the dynamic state $\Delta\mathbf{k}$. All processes – absorption, emission, exchange, pair annihilation, pair creation – are systematically described. The relative sign of t-axis components determines attraction/repulsion (S4).

4. **Strong interaction**: Gluons mediate Q-axis transitions and exchange color charge. Absorption, emission, pair annihilation, pair creation are systematically described. The color neutrality condition provides the origin of confinement, and asymptotic freedom holds inside the set (S5).

5. **Weak interaction**: $W^{\pm}$ mediates t-axis displacement (flavor conversion). In the quark system, this acts as a sign inversion of k_t; in the lepton system, the simultaneous satisfaction of t-axis displacement and $\kappa = 2$ preservation produces axis reconfiguration between charged leptons and neutrinos. In addition to absorption and emission, W decay and inverse decay are unified as Type D/C. $k_t \neq 0$ provides the origin of CP violation and the left-handed chirality selection rule (S6).

6. **Gravitational interaction**: The graviton acts on both (t, R) axes and couples to all waves with $\Delta\mathbf{k} \neq 0$ (universal coupling). Absorption and emission are systematically described (S7).

7. **Origin of mass**: Coupling with the standing wave ($\kappa = 1$, z axis) reduces the speed ratio $|k_{s_0}/k_t|$ of direction-constrained waves, producing mass as $v < c$. The 3rd generation, which directly shares the z axis, has the strongest coupling strength, providing the origin of the inter-generation mass hierarchy $m_3 \gg m_{1,2}$. Spherical waves with $\kappa_s = 0$ lack a coupling axis and have zero mass (S8).

8. **Wave packet deformation and causal structure**: Wave packet collapse is a conservation-law-governed deformation and "observation" is unnecessary. Interactions mediated by spherical waves with $k_t = 0$ are causally symmetric, and the path, localization, and variational principle are retroactively constructed. The case $k_t \neq 0$ has causal temporal ordering (S9).

The correspondence with Table A of [4] is shown below:

Interaction	Spherical wave	Non-zero axis	Effect on direction-constrained wave	k_t	Causality	Fundamental vertices
Higgs coupling	Standing wave H^0	z axis	Decrease in v (mass)	0	–	–
Electromagnetic	γ	R	$\Delta\mathbf{k}$ change (scattering)	0	Retroactive	A, B, C, D
Strong	g	Q	Q change (color exchange)	0	Retroactive	A, B, C, D
Weak	$W^{\pm}$	t	k_t change (flavor conversion)	$\neq 0$	Causal	A, B, C, D
Gravity	G	t, R	$\Delta\mathbf{k}$ change (acceleration)	$\neq 0$	Causal	A, B

All claims in this paper are stated as geometric facts, and the physical correspondences are presented as "connectable interpretations."

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